

Summary

Employee Attrition Prediction using Machine Learning

Overview :

The purpose of this project was to identify the factors that influence employee attrition and provide insights that can help the Human Resources (HR) team reduce employee turnover. By analyzing employee information such as salary, department, job role, work-life balance, years at the company, and other workplace factors, we identified patterns that are associated with employees leaving the organization.

Key Findings :

The analysis showed that employee attrition is not evenly distributed across the company. The Sales department experienced the highest employee turnover, with Sales Representatives leaving more frequently than employees in other roles. Most resignations occurred during the first few years of employment, indicating that retaining new employees should be a priority. Employees with lower monthly income were more likely to leave the company. However, salary was not the only reason for attrition. Factors such as work-life balance, job satisfaction, overtime, and length of service also played an important role in an employee's decision to stay or leave.

The study also found that employees reporting poor work-life balance had a noticeably higher likelihood of leaving the organization. These findings suggest that improving the overall employee experience is just as important as offering competitive compensation.

Business Recommendations :

Based on these findings, HR should focus retention efforts on employees working in the Sales department, particularly Sales Representatives and employees in their early years with the company. Regular one-on-one discussions, career development opportunities, mentoring programs, and stronger onboarding can help improve employee engagement and reduce early resignations. The company should also review compensation for employees in lower salary ranges while continuing to improve work-life balance through flexible work arrangements, workload management, and employee wellness initiatives. These actions can help increase employee satisfaction and reduce unnecessary turnover.

Conclusion :

This project demonstrates that employee attrition is influenced by multiple workplace factors rather than a single reason. By identifying employees who may be at a higher risk of leaving, HR can take proactive steps to improve employee satisfaction, strengthen retention strategies, and reduce the costs associated with hiring and training new employees. Although the analysis provides valuable insights, it should be used as a decision-support tool alongside HR expertise, since personal and external factors affecting employee decisions are not captured in the available data.

By :

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